

Connectionless Transport: UDP

1 Introduction

User Datagram Protocol (UDP), defined in **RFC 768**, is the simplest transport protocol built on top of IP. It is often called a **connectionless transport protocol** because it does not establish or maintain connections between sender and receiver.

Key Features of UDP

- **Connectionless:** No handshaking before communication.
- **Lightweight:** Only multiplexing/demultiplexing and error checking.
- **Unreliable:** No guarantee of delivery, order, or duplicate suppression.
- **Minimal Overhead:** Only 8 bytes of header.

2 Why UDP?

Unlike TCP, UDP provides a **bare-bones service**. Applications choose UDP over TCP when they need:

1. **No connection establishment:** Avoids TCP's three-way handshake, eliminating RTT delays.
2. **Stateless operation:** No connection state at sender or receiver, allowing high scalability.
3. **Small header size:** Only 8 bytes, minimizing overhead compared to TCP's 20 bytes.
4. **No congestion control:** UDP can send as fast as desired, even during network congestion.
5. **Low-latency friendly:** Ideal for real-time applications (VoIP, gaming, streaming) that tolerate some loss over delay.

3 How UDP Works

3.1 Sending Side

1. Application passes message to UDP.
2. UDP attaches header (source/destination ports, length, checksum).
3. Segment is handed to IP for delivery.

3.2 Receiving Side

1. IP delivers UDP segment to transport layer.
2. UDP verifies checksum (optional in some cases).
3. Based on the **destination port**, message is delivered to the correct application process.

3.3 Connectionless Nature

There is **no handshaking** between sender and receiver. Each segment is independent, like sending individual postcards.

4 Applications Using UDP

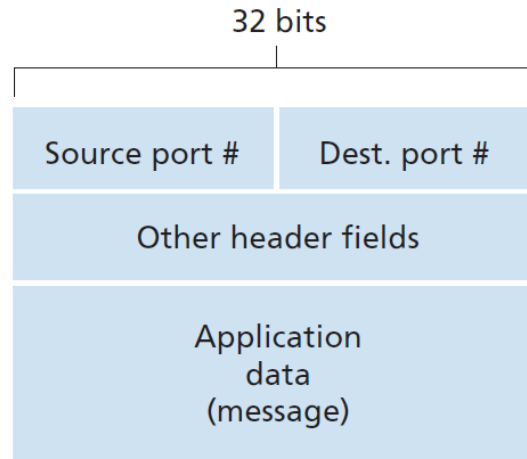
- **Streaming Multimedia Apps:** Used for applications like VoIP and video conferencing because they are **loss tolerant** but **rate sensitive** (prefer speed and low delay over perfect reliability).
- **DNS (Domain Name System):** Fast queries without the overhead of connection setup.
- **SNMP (Simple Network Management Protocol):** Works efficiently, even under stressed or busy networks.
- **HTTP/3:** The latest version of the Hypertext Transfer Protocol uses a UDP-based transport layer (QUIC) for reduced connection latency.
- **Gaming & Real-time Apps:** Prioritize speed over reliability.

Reliable Transfer Over UDP

- If **reliable transfer** is needed over UDP (e.g., for HTTP/3, which uses QUIC):
 - The needed **reliability** (e.g., retransmission, ordering) is added at the **application layer** (or in the protocol layer built on top of UDP, like QUIC).
 - **Congestion control** mechanisms are also added at the **application layer** to manage data transfer rates responsibly.

5 UDP Segment Structure

UDP header is only 8 bytes:



Source Port (16)	Destination Port (16)
Length (16)	Checksum (16)
Application Data (Variable length)	

- **Source Port:** Identifies sending process.
- **Destination Port:** Identifies receiving process.
- **Length:** Total length of UDP segment (header + data).
- **Checksum:** Error detection across segment + parts of IP header.

6 UDP Checksum

The checksum provides end-to-end error detection:

1. Divide the segment into 16-bit words.
2. Compute the **1's complement sum** of all words.
3. Take the **1's complement** of the sum to get the checksum.

Example

Suppose we have three 16-bit words in a UDP segment:

Word 1: 1010 1100 1111 0000 Word 2: 0101 1010 0000 1111 Word 3: 1111 0000 1010 1010

1. Compute 1's complement sum:

$$1010110011110000 + 0101101000001111 = 1111011011111111$$

$$1111011011111111+1111000010101010 = 1110011100101001 \quad (\text{wraparound if needed})$$

2. Take 1's complement of the sum:

$$\text{Checksum} = 0001100011010110$$

At the receiver:

- Add all words including checksum.
- If result = 1111...111, no error.
- Otherwise, error detected.

End-to-End Principle

Since not all links provide error detection, UDP includes its own checksum to ensure **end-to-end reliability check**. This demonstrates the **end-to-end principle**: certain functions must be implemented at the endpoints, not intermediate nodes.

7 Summary

UDP offers:

- Minimum functionality (multiplexing, demultiplexing, and error detection).
- No guarantees for reliability, ordering, or congestion control.
- Suitable for **real-time, low-latency, or scalable** applications.

TCP, in contrast, provides reliability and congestion control at the cost of higher complexity and overhead. UDP enables applications to implement their own reliability mechanisms if needed (e.g., QUIC over UDP).